Central Tendency

Computational Statistics Team Teaching

2024/2025

Outlines

- What is central tendency?
- How to measure central tendency?

What is central tendency in statistics?

Resume of the data

Central Tendency 🎯

Definition

- A statistical measure that identifies a single value as representative of an entire dataset
- It aims to describe a dataset by finding the central or typical value around which the data points are distributed.
- The three most common measures of central tendency are mean, median, and modus (*mode*)

Measurement of Central Tendency

How to measure central tendency?

Measurement of Central Tendency

Central Tendency

- **Mean**: Arithmetic average
- **Median**: Middle value
- **Mode**: Most frequent value

Mean → Arithmetic Average

Mean

The mean is calculated by **summing all the values** in the dataset and **dividing by the total** number of observations.

Formula

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \cdots + x_n}{n}$$

Formula

$$\bar{x} = \frac{\sum x}{n}$$

Mean → Sample vs. Population

Sample

$$\bar{x} = \frac{\sum x}{n}$$

X bar

Population

$$\mu = \frac{\sum x}{n}$$

mu

Mean → Simple Case (1)

Kasus

A basket contains 10 apples with the following individual weights: 100g, 200g, 150g, 100g, 120g, 80g, 90g, 160g, 110g, 170g.

Mean

$$\mu = \frac{100 + 200 + 150 + 100 + 120 + 80 + 90 + 160 + 110 + 170}{10} = \frac{1280}{10} = 128$$

Mean → Simple Case – Code Implementation

Without Library

```python
# Buat daftar buah apel dalam bentuk list
apel = [100, 200, 150, 100, 120, 80, 90, 160, 110, 170]

total_berat = 0 # variabel untuk menyimpan total berat seluruh apel

for i in range(len(apel)):
    total_berat = total_berat + apel[i] # jumlahkan semua berat apel

avg_apel = total_berat / len(apel)
print(avg_apel)
```

Using Numpy

```python
# Dengan Numpy
import numpy as np
apel = [100, 200, 150, 100, 120, 80, 90, 160, 110, 170]
avg_apel = np.mean(apel)
print(avg_apel)
```

Median → Middle Value

Nilai Median

The median is the middle value of a dataset when it is **arranged in ascending order**.

Example

20gr, 30gr, 100gr, 50gr, 40gr → 20gr 30gr, 40gr, 50gr, 100gr

20gr 30gr, **40gr**, 50gr, 100gr

Median → Odd vs. Even Observations

Odd

20gr, 30gr, 100gr, 50gr, 40gr → 20gr 30gr, 40gr, 50gr, 100gr

20gr 30gr, **40gr**, 50gr, 100gr

Even

20gr 30gr, **40gr**, **45gr**, 50gr, 100gr

$$median = \frac{40 + 45}{2} = 47.5$$

Median → Simple Case Code Implementation

Without Library

```python
import math

def bubblesort(elements):
    for n in range(len(elements)-1, 0, -1):
        for i in range(n):
            if elements[i] > elements[i + 1]:
                elements[i], elements[i + 1] = elements[i + 1], elements[i]
    return elements


def calc_median(elements):
    elements = bubblesort(elements)
    len_element = len(elements)
    if len_element % 2 is 0:
        center = math.floor(len_element/2)
        return (elements[center-1] + elements[center]) / 2
    else:
        return elements[math.ceil(len_element/2)-1]

apel = [100, 200, 150, 100, 120, 80, 90, 160, 110, 170]
median = calc_median(apel)
print(median)
```

Using Numpy

```python
import numpy as np
apel = [100, 200, 150, 100, 120, 80, 90, 160, 110, 170]
med_apel = np.median(apel)
apel.sort()
apel_urut = ' '.join(map(str, apel))
print('Apel urut : {:s}'.format(apel_urut))
print(med_apel)
```

Mode → Most Frequent Value(s)

Nilai Modus

The mode is the value that appears most frequently in the dataset.

Example

Student score in Class XII SMA XYZ → 70, 80, 70, 70, 90, 100

Score 70 → 3 data; Score 80 → 1 data; Score 90 → 1 data; Score 100 → 1 data

Mode = 70

Mode – The Value

Mode in Obeservations

- No Mode
- One Mode
- Multiple Mode

Modus → Simple Case in Code

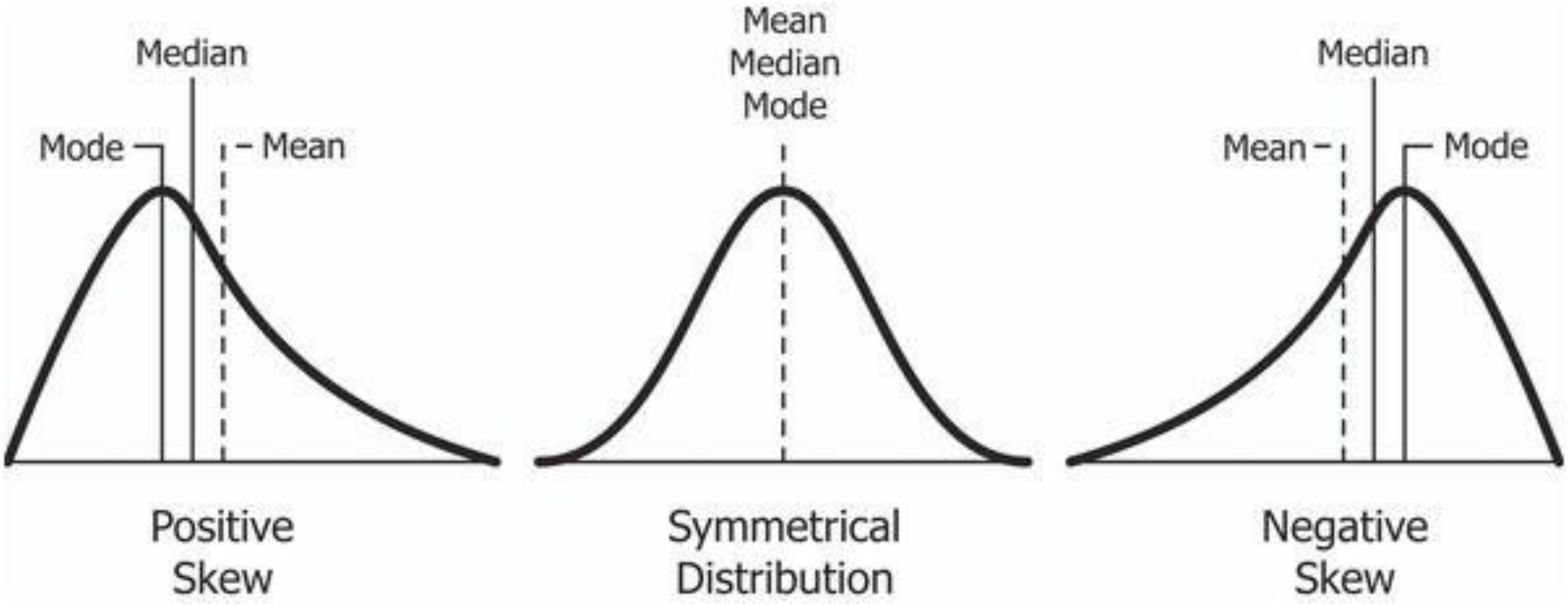
Code Using Scipy vs. Native

```python
from scipy import stats
import statistics as s

nilai = [70, 80, 70, 70, 90, 100]

mode_nilai = stats.mode(nilai) #scipy
modus = s.mode(nilai) # native python
print(modus)
print(mode_nilai)
print(mode_nilai.mode)
```

The Benefit of Central Tendency Measurement

What is the benefit of measuring central tendency?

Considering the skewness in observations



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